

Integrating Synchrophasor Technology into Power System Protection Applications

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Abstract—This paper analyzes the potential for using synchrophasors for protection system applications. This paper offers a review of synchrophasor data communication, latency and timing as well as an analysis of protection schemes including differential, step distance, out-of-step and oscillatory stability protection and the potential for using Phasor Measurement Unit (PMU) data.

Index Terms-- Phasor Measurement Unit, Relays, Protection System

I. INTRODUCTION

This paper analyzes the potential for using synchrophasors for protection system applications. The paper first reviews the current state of Phasor Measurement Unit (PMU) data collection and communications and compares that with the current data collection standards of protective relay systems. Second, the paper offers a general analysis of representative protection schemes and whether and how each could use synchrophasor data. The paper addresses:

- Analysis of protection system data processing timing as compared to PMU data collection timing to determine where and how protection systems can benefit from the use of synchrophasor data,
- A review of protection system applications and how PMUs can be applied, and,
- An analysis of PMU data latency and security.

II. PROTECTION SYSTEMS VERSUS SYNCHROPHASOR NETWORK/DATA

System protection for the bulk power system entails the use of hardware, software, planning rules and standards to prevent cascading outages, protect power grid equipment and maintain system reliability. To date, system protection has primarily relied upon protective relaying, used within substation or local area communication-assisted systems. With the availability of high-quality, high-speed

communications networks, protection-oriented relaying systems now span across Transmission, Distribution, and Generation. Recent power system performance standards have required expansion of protection systems to cover wider areas using Remedial Action Schemes (RAS) with multiple relay communication networks; these require highly accurate, reliable, real-time system-wide data.

System protection departments in utilities are responsible for developing and implementing protection schemes and programming power system relays to operate, trip and reclose high voltage equipment to isolate fault currents or mitigate potential system contingencies. The challenge is to balance protecting system equipment from damage while ensuring reliability by reducing outage duration.

Every protection scheme operates by examining a number of signals or devices that bound the region or device to be protected. For most protection functions (such as a distance function aligning voltages and currents, or a synchro-check function comparing voltages) all of those measurements from devices at different physical locations must be synchronized to an absolute time. Furthermore, most system protection schemes in use today are triggered based upon whether conditions at specific points on the grid reach pre-determined minimum or maximum levels – even though system topology changes constantly and those trigger levels may not be problematic for a wide range of dynamic system configurations. The benefit of using time-stamped PMU data is that it allows protection to be set based on actual grid stresses as measured by phase angles in real time, producing more appropriate protection operations,¹ and the time-stamped data can be time-aligned automatically for analysis.

There are three key questions in assessing the potential for using synchrophasor data in lieu of relay measurements for

¹ O'Brien, J., A. Deronja, et al., "Use of Synchrophasor Measurements in Protective Relaying Applications," Power System Relaying Committee, Report of Working Group C-14 of the System Protection Subcommittee, April 8, 2013, p. 5.

the information needed to determine whether to trigger and actuate a regular protection scheme:

- 1) Do PMU-collected data offer an advantage in terms of the types, measurement speed and accuracy of grid information collected, relative to current relay data used in existing protection systems?
- 2) Do current synchrophasor data networks move PMU data quickly enough to meet local or system protection timing requirements, or can dedicated PMU data transfer measures be established (much as there are dedicated data lines for many relay-based system protection schemes today)?
- 3) Where relays are connected to dedicated communications lines to execute a specific protection scheme, can those existing lines be used to deliver PMU data as well? This may be feasible particularly where the PMUs are not stand-alone devices, but relays that have been converted to exploit built-in PMU functionality. This could allow the time-stamped PMU data to supplement or replace the relay data in protection scheme assessments and execution.

The first question goes to whether PMU data measurement speeds (at 30 up to 120 samples per second) and calculations (such as phase angles) are more useful than relay measurements for particular types of protection schemes. This is clearly an application-dependent issue – some protection uses such as differential and distance protection need very fast grid condition measurements, while others can use directly calculated phase angles in the decision on whether to trigger the protective action. This question will be addressed in Section IV and V.

III. PMU DATA DELIVERY

The second question addresses whether the measured grid data can be delivered to the point of protection analysis and actuation fast enough to meet the protection system's requirements. This question concerns both how quickly existing relay protection data reach the point of analysis and tripping, and whether PMU data can meet or exceed those speeds. To frame this issue, this paper proposes that it will be worthwhile to study the speed of existing relay protection analog signal capture and logical tripping speeds, and the current speed of PMU analog signal capture and delivery. As an example of the first type of study, in 2013 Salt River Project analyzed the tripping times of two vendor relays for three types of protection systems: line protection, transformer protection and bus protection. The test set-up included:

- Direct back-to-back fiber connections for relays requiring communication,
- Analog signals provided in series between the two relays being tested (two tests, each with the vendor CT signals swapping spots in the series string),
- Relay logic providing no additional delay in trip signals, and,

- A third party measurement of timing results to prevent bias.

The results are shown in Table 1.

TABLE 1
Relay Trip Timing Tests

Line Relaying Distance	Test Number	Test Description	Time difference (cyc)	First to trip
	1A	Local distance test, Vendor 1 1st in CT string, Communication disabled	1.2	VENDOR 2
	1B	Local distance test, Vendor 2 1st in CT string, Communication disabled	1.34	VENDOR 2
Line Relaying Differential	2A	Differential test for Direct Transfer Trip, Vendor 1 1st in local CT string, Vendor 2 first in remote CT string	2.69	VENDOR 2
	2B	Differential test for Direct Transfer Trip, Vendor 2 1st in local CT string, Vendor 1 first in remote CT string	2.45	VENDOR 2
	3A	Differential test, Vendor 1 1st in local CT string, Vendor 2 first in remote CT string	0.144	VENDOR 1
	3B	Differential test, Vendor 2 1st in local CT string, Vendor 1 first in remote CT string	0.336	VENDOR 1
Transformer Relaying Differential	1A	Differential test, Vendor 1 1st on high side winding, Vendor 2 1st on low side winding	0.456	VENDOR 2
	1B	Differential test, Vendor 2 1st on high side winding, Vendor 1 1st on low side winding	0.456	VENDOR 2
	2A	Overcurrent test, Vendor 1 1st on high side winding, Vendor 2 1st on low side winding	0.528	VENDOR 1
	2B	Overcurrent test, Vendor 2 1st on high side winding, Vendor 1 1st on low side winding	0.528	VENDOR 1
Bus Relaying	1A	Differential test, Vendor 1 1st on first bus CT winding, Vendor 2 1st on second bus CT winding	0.096	VENDOR 1

With the understanding of this type of timing data, PMUs can be tested for their analog input and data delivery speed to provide a comparison to understand the architecture required for a PMU-based protection network.

Current synchrophasor data networks have been designed principally to collect data from substation-based PMUs and deliver the data up to regional Phasor Data Concentrators (PDCs) for centralized wide-area monitoring, visualization and analysis. Most current PMU signal networks have three basic structures: PMU to PMU (Fig. 1), PMU to PDC to PMU (Fig. 2), and PMU to PDC to Central Logic Processor (Fig. 3).

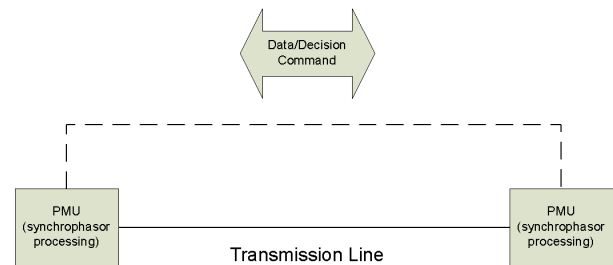


Figure 1. PMU to PMU Network Architecture

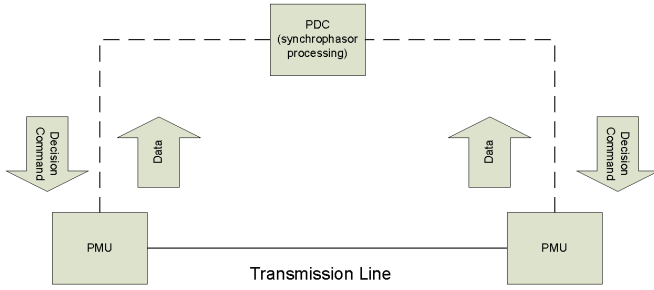


Figure 2. PMU to PDC to PMU Network Architecture

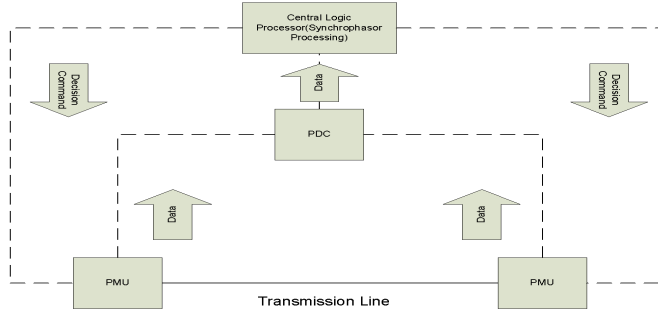


Figure 3. PMU to PDC to Central Logic Processor Network Architecture

It is within these sample synchrophasor network structures that the data capture and communication rates of potential PMU protection systems should be analyzed. While the studies proposed above will prove the usability of synchrophasors for protection, the quality of the data as it travels through the network structure is of vital importance. For, if PMU data can be usable for protection, the effect on data quality as it passes through the mediums present in the network structure needs to be quantified and understood.

But at the same time, it is worth noting that current system protection schemes often incorporate dedicated communications lines to deliver relay data to the point of protection analysis and decision within the protection scheme's required timeline (as compared to only using the utility's SCADA on general data networks). It is therefore inappropriate to assume that PMU data cannot be used for system protection because the latency costs on the primary PMU communications path from the field PMU to the central PDC and back to the protection point are too high. It may be possible to build comparable dedicated data lines or to run PMU data across the existing protection system communications lines if there is a protection benefit to doing so.

IV. SYNCHROPHASORS IN POWER SYSTEM PROTECTION

This section provides an overview of a representative example of protection schemes and their application with synchrophasors including distance protection, differential protection, out-of-step protection and oscillatory stability protection. Other protection schemes have been analyzed as

part of this effort and can be viewed in the full NASPI EATT Report. [9]

A. Distance Protection

Every transmission line, as seen from one end, has a characteristic impedance that represents the line length based on Ohms law rearranged for impedance, $Z=V/I$. The calculated effective load impedance from a fault using fault current and pre-fault voltage is compared to the expected characteristic line impedance to determine fault location.

Distance relaying also allows for remote line backup protection by incorporating zones of protection. Zone 1 (Z1) is used for the local line protection and is usually set for 80% of the line length to allow for uncertainties in the actual line impedance. Zone 2 (Z2) is used to protect the next forward connected lines from the local relay on a time delay, usually set for 20 cycles, and is set to "see" the entire local line length and an additional percentage into the next line. By setting the impedance reach higher for this zone, overlapping protection can be accomplished between transmission lines. This same concept is applied to Zone 3 (Z3) protection to see two lines out. Fig. 4 below displays this concept as the line relay at bus G protecting line XR can provide backup protection for a portion (typically 50%) of line YQ on time delay T2 in case the relay at bus H fails, and a portion (typically 10%) of line ZP on time delay T3 in case the relay at bus R fails.

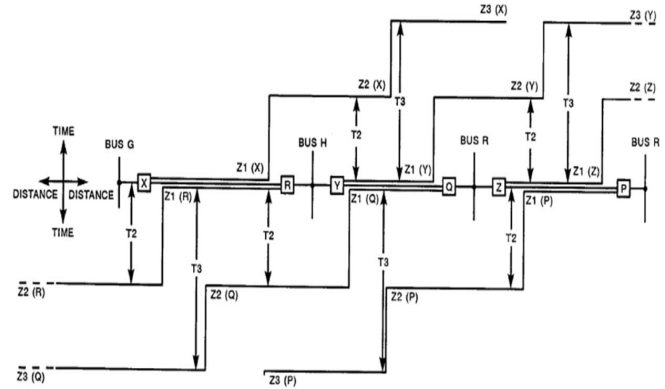


Figure 4. Step Distance Zone protection

Application of PMUs -- If multiple distance elements are implemented in various relays over a wide area, a multi-zone pilot-operated scheme could be developed incorporating PMU data from several PMUs. Implementing this type of scheme within relay logic is typically either too cumbersome or possibly unachievable without a dedicated remedial action scheme; however, the same scheme could be implemented on a synchrophasor-capable programmable logic controller (PLC) to perform the required protection computation and coordination actions. Current P-class PMU technology today does report phasor data at rates fast enough for coordinated backup line distance protection. The phasor calculation latency for protection-rated PMUs is shorter than the delayed

pickup durations of typical zone 3 distance protection configurations, so this protection use is fully feasible.

B. Differential Protection

Differential protection basic theory is based on Kirchhoff's current law for electrical circuits: The sum of all currents entering a node must equal zero. This law can be restated as the sum of all currents entering a node must equal the sum of all currents exiting that node. In power system terms, this law translates as any current entering one electrical terminal of any protected device must exit the opposite electrical terminal of that device. The difference in current angles on each end of the device should be close to 180 degrees with small variance due to loading conditions. In a faulted condition, current will flow to the path of least resistance, to the fault on the device, and thus the current direction, specified by the angle, at both terminals will be equal to 0 degrees. Differential protection theory is applied to transmission lines, transformers, bus work and generators. Fig. 5 graphically depicts transmission line current differential theory.

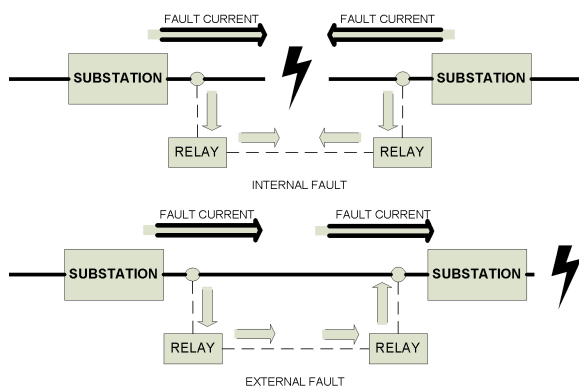


Figure 5. Transmission Line Differential Protection Theory of Operation

To account for potential current transformer saturation issues, analytics use a percent differential which calculates the ratio of the summation discussed above with a phasor magnitude average. (Fig. 6)

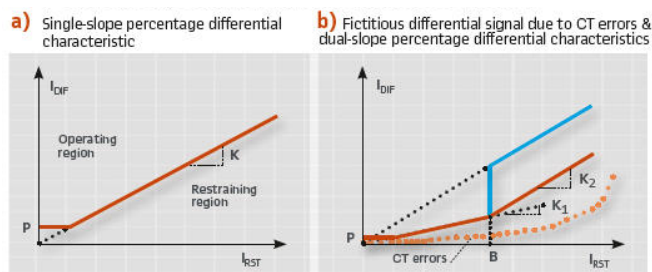


Figure 6. Percent Differential Protection²

² Pacworld, "Adaptive Line Differential Protection," June 2012: https://www.pacworld.org/nocache/issue/june_2012_issue/adaptive_line_differential_protection/adaptive_line_differential_protection.html

As demonstrated in Figure 6b above, if a CT saturates and misrepresents the actual current on the line, the restraining current, I_{RST} , on the X axis, will increase greatly while the differential current, on the Y axis, I_{DIFF} , will increase to a lesser degree. This will place the percent differential calculation in the restraining region of the graph and block operation.

Application of PMUs -- PMUs can be used for line differential protection because current phasors are measured and can be transmitted to a central or remote end(s) of the line for the differential calculation. PMU measurements are time-synchronized and therefore provide a common time reference for aligning the phasor measurements from each end of the line being protected. Each PMU serving the zone of protection can calculate the differential current as the sum of currents entering the line, in every time-stamped set of measurements.

Line differential protection calculations should be performed in a sub-cycle timeframe; therefore, PMUs would need to measure phasor data at rates of at least 2 samples per cycle (120 samples per second), or preferably 4+ samples per cycle. However, the quality of PMU data at these rates will need to be qualified as reliable for protection purposes. Current state-of-the-art PMUs on the market today report phasor data at a maximum of 120 samples per second, or 2 samples per cycle; however, most PMUs deployed today are reporting at rates of 30 or 60 samples per second and would not be sufficient for line differential protection. There are additional latency costs incurred as the protection measurements are transmitted to the remote terminals; unless the EHV bus where PMUs are installed has a high-bandwidth, low latency communications channel to deliver the data from the PMU to the remote terminal, the protection calculation and action may be delayed for too long to be effective. This type of protection requires minimal latency and would generally require direct PMU-to-PMU communications without passage through a PDC; similar dedicated local relay-to-relay communications links are routine protection measures today.

Bad PMU data or loss of time synchronization for one of the PMUs measuring the terminal current poses a risk for this application in terms of false tripping (as does loss of timing for a protection relay). Loss of synchronization manifests into an incorrect phase angle value that compromises the current phasor used in the differential protection calculation.

PMUs have been proposed in [3] as a backup transmission line negative-sequence current differential element. The performance of this system improves with faster messaging rates and filtering processes.

C. Out of Step Protection

Out of Step protection involves being able to either separate generation when slip frequency and the acceleration of slip frequency drives the system in an unstable position, or blocking tripping during a power system swing that can correct itself without separating the system. Fig. 7 illustrates

power swing regions with respect to slip frequencies and acceleration.

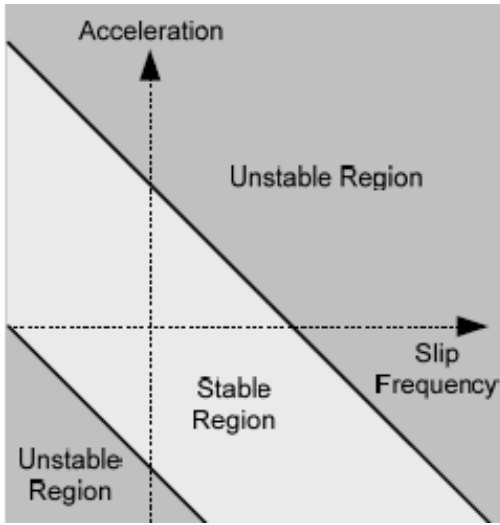


Figure 7. Out of Step Stability Graph

Generator out-of-step protection trips an unstable generator losing synchronism with the grid. Loss of synchronism conditions causes high stator winding currents, mechanical stress, pulsating torques, and resonance effects that can result in severe damage to the generator. The most usual form of out-of-step protection uses a distance relay function to monitor the impedance trajectory seen at the terminals of the generator. Fig. 8 shows an equivalent system and the equivalent impedance seen by the relay using the voltage and current phasors.

$$Z = \frac{V}{I} = \frac{E_G \angle \delta - I \cdot j \cdot (x_G + x_T)}{E_G \angle \delta - E_S} = -j \cdot (x_G + x_T) + j \cdot (x_G + x_T + x_s) \cdot \frac{E_G \angle \delta}{E_G \angle \delta - E_S}$$

$$= j \cdot \frac{(x_G + x_T + x_s) \cdot n \cdot (n - \cos \delta - j \cdot \sin \delta)}{(n - \cos \delta)^2 + \sin^2 \delta} - j \cdot (x_G + x_T)$$

$$n = \frac{E_G}{E_S}$$

Figure 8. Example Two-Source System

The impedance locus is illustrated in Fig. 9. For $n = 1$, the impedance trajectory is a straight line that is perpendicular to and crosses the midpoint of the total impedance $Z_G + Z_T + Z_S$. For $n \neq 1$, the trajectory is a circle.

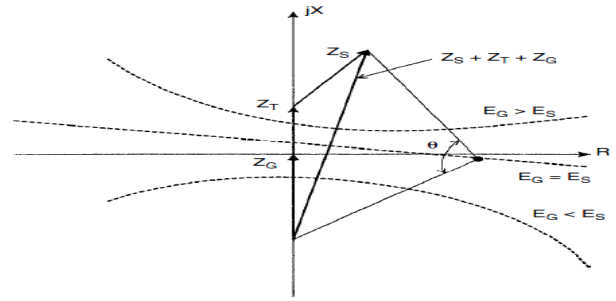


Figure 9. Impedance Trajectory during a Power Swing [1]

Common out-of-step protection schemes use a mho relay function with single or double blinders, as shown in Fig. 10. For single blinder protection, out-of-step conditions are detected if the impedance trajectory remains within the two blinders for a time greater than a specified duration, or if it crosses both the blinders. For example, scenarios "a" and "b" are considered stable scenarios while scenario "c" is an unstable scenario. Double blinder schemes use a timer between entering the outer blinder (RRO) and inner blinder (RRI) and tripping occurs if the timer expires, identifying an unstable swing condition.

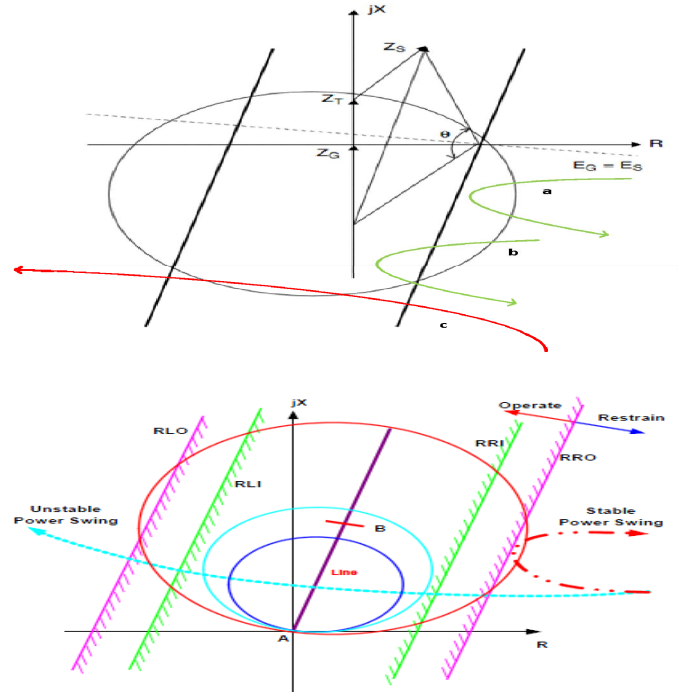


Figure 10. Single and double blinder out-of-step protection schemes [2]

Application of PMUs -- A novel predictive generator out-of-step protection scheme has been developed which utilizes synchrophasor technology. The generator out-of-step protection scheme is enabled by a synchrophasor-based dynamic state estimator, along with a measurement based dynamic equivalencing technique for deriving in real time an updated dynamic equivalent of the system. This equivalent is used for the stability characterization of the generator with respect to the system based on Lyapunov's direct method.

The distributed dynamic state estimation is performed in a substation utilizing synchronized and non-synchronized local measurements and provides a highly accurate and verified dynamic model of the substation and incoming power circuits. For a substation which serves a generator, the scheme can use this data scheme to identify the center of oscillations of the system.

This procedure is initiated when a disturbance is detected. Given the center of oscillations, an equivalent system is derived that mimics the dynamics of the actual system. The equivalent dynamic system is updated continuously and is utilized for the characterization of the generator stability. Specifically, the total energy of the generator (potential plus kinetic energy) is continuously monitored and compared to the stability limit of the total energy function. When the total energy exceeds the stability limit then instability is asserted and a trip signal is sent to the generator breaker. The major advantage of the proposed scheme is that the out-of-step condition is predicted before its occurrence, when the angle between the generator and the system is relatively small. In this case the generator can be tripped much faster than the traditional out-of-step technology for the simple reason that breaker transient recovery voltage (TRV) phenomena will be mild. The generator out-of-step protection scheme has been demonstrated through simulations on a generating substation of New York Power Authority's (NYPA) system and was compared with the response of a conventional impedance out-of-step relay. For a simulated unstable scenario the developed method predicted instability 0.2 seconds before the conventional relay, enabling action before the generator slipped a pole and the breaker was not overstressed with transient recovery voltage. [5]

D. Oscillatory Stability Protection

Monitoring and protecting against undamped oscillatory behavior on the power system is a valuable use of synchrophasor technology for protection and control. For example, the Guatemala power system is connected to the 50 GW grid in Mexico and the 7GW grid of the remaining Central American countries. When it observes sustained, undamped oscillatory behavior across its interconnection, Guatemala has deployed a real-time stability protection scheme using PMU measurements that monitors for undamped oscillatory conditions and separates from the smaller 7GW Central American system, remaining interconnected with the stiffer Mexico power grid. [6] Fig. 11 shows the electrical topology and an example separation event detected by PMUs resulting in automated separation.

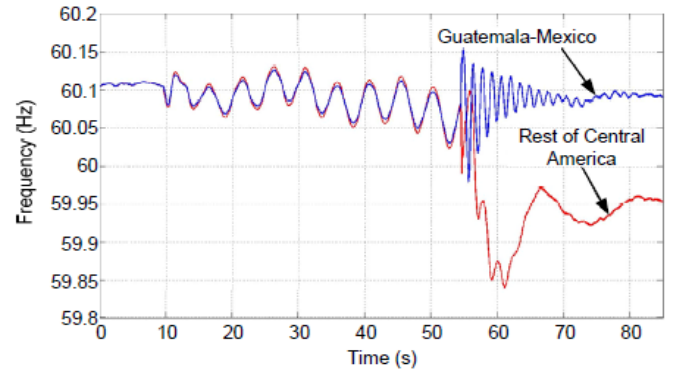


Figure 11. Guatemalan Power System and PMU-Based Stability Scheme [6]

The synchrophasor-based monitoring system detects oscillations on the real power flow measurements across the system. Modal analysis (modified Prony analysis) is implemented on the SEL-3378 Synchrophasor Vector Processor. Voltage and current phasor measurements are reported 30 times per second via C37.118 protocol. Negative damping ratio triggers the Remedial Action Scheme (RAS), separating the tie lines connecting Guatemala to the rest of Central America.

Application of PMUs -- Oscillation frequencies for local and inter-area oscillations are relatively low (< 5 Hz) and are accurately measurable by PMUs reporting at 30 or 60 samples per second. It has been proven that PMUs can detect undamped growing oscillations either in a pre-contingency or post-contingency condition in actual system operations. Marginally damped or slightly negatively damped oscillations will manifest over a relatively long period of time. This allows for automated controls or even system operators to take action upon robust detection of the unstable oscillations occurring. While this solution may be system-dependent, PMUs may be an inexpensive and sustainable tool to inform and operate remedial actions to mitigate these.

V. SIGNAL DELAY – SYNCHROPHASOR SIGNAL LATENCY

Signal latency is a prime concern for the application of synchrophasors in protection systems. From section 5.3.4 of the IEEE C37.118-1 standard:

...PMU reporting latency is defined as the maximum time interval between the data report time as indicated by the data time stamp, and the time when the data becomes available at the PMU output (denoted by the first transition of the first bit of the output message at the communication interface point) [7].

In addition to this internal device latency, “reporting over network delay” is an even bigger factor including communication over varying distances. The standard reports in section 5.5.9 that the output latency should be $2/F_s$ (sampling/reporting frequency) for Protection class data and $7/F_s$ for Metering class data. Therefore, P-class synchrophasor data has lower latency in order to achieve high

speed reporting, whereas M-class PMUs use more filtering to produce more accurate measurements, but have a higher latency. This is the scope of the IEEE standard's current recommendations for synchrophasor latency.

Most synchrophasor data networks have been designed to flow data up from field PMUs to Transmission Owner and Reliability Coordinator control rooms for centralized monitoring and analysis and action. However, there is no reason why specialized, local, low-latency communications structures cannot be designed and built to exploit the benefit of PMU-based measurements for system protection, particularly at edge applications. Since there already are dedicated communications supporting particular SPS and RAS operations, similar communications could be deployed to support PMU-based SPS and RAS schemes.

If current synchrophasor communications networks are used for system protection, the synchrophasor signal needs to pass through the PMU, communication networks, and possibly one or multiple levels of Phasor Data Concentrators (PDCs), each of which introduces some additional signal latency. Here is one calculation of the latency impacts of multiple levels of PDCs [8]:

- The substation PDC is installed at high voltage substations, and can also be connected to other, surrounding substation PMUs...[with] low internal latency (around 3 to 10 ms).
- The regional PDC is installed at regional control and operating centers (if a utility has this level in their power delivery hierarchy) ... [with] moderate internal latency (approx. 10 to 100ms).
- Super PDCs are normally installed at the main grid control center, where the EMS of the grid is deployed.... [and] possess[es] moderate internal latency (approx. 100 ms to 1 sec).

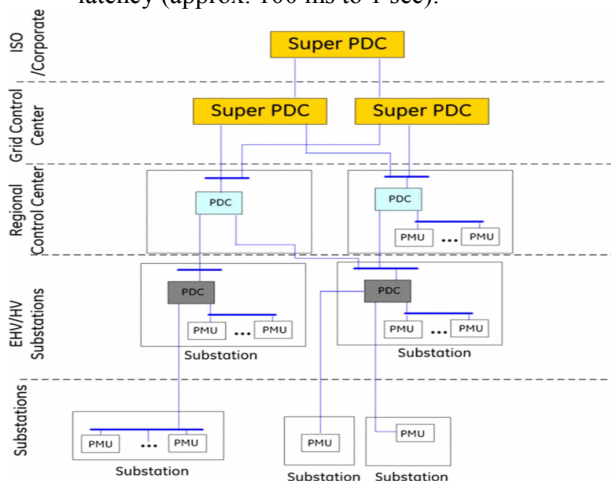


Figure 12. Wide Area Measurement System (WAMS) hierarchy [8]

Different PDC architectures have different latency costs. [8]:

The centralized WAMS [Wide Area Measurement System] architecture eliminates intermediate levels between control centers and the substation... [with PMUs] connected to a substation PDC and

then to Super PDC or directly to Super PDCs.

This creates low latency due to the reduced number of intermediate nodes, but drastically increases the number of PMU signals to the Super PDC. More intermediate PDCs create higher signal latency, but more data-concentrating power. There are also differing impacts from hardware and software PDCs - while embedded hardware PDC latency is well determined, the latency of software PDCs across different physical hardware and operating systems is less established. A hybrid of these methods should be determined based on the individual PMU application requirements. One conclusion is "a hard-real time platform with dedicated embedded system hardware can reduce internal PDC latencies, and increases the throughput for the time-critical applications." [8] Latency could be further improved with emerging communications network architectures intended to serve edge applications for synchrophasor technology and decentralized system protection.

VI. SIGNAL SECURITY – LOSS OF GPS TIME SYNCHRONIZATION

Accurate timing is vitally important for PMUs because correct operation of a PMU requires a common and accurate timing reference available to all PMUs across a power system. Within a PMU, a phase-locked oscillator is used to generate the time tags within the second. The time tag is sent out with the phasors in a data packet. Thus if a phasor information packet arrives out of order to a phasor data concentrator (PDC), the phasor time response can still be assembled correctly. If the time signal (e.g., GPS pulse) is delayed, then it could cause a time-tagging error and lead to the calculation of a significant phase error.

The time system of a PMU-based GPS system is depicted in Fig. 13. The standard temporal reference of this system is generated with a signal of one pulse per second (one PPS) from a GPS. This pulse as received by any receiver on earth is coincident with all other received pulses to within 1 microsecond. The PPS signal is used for sampling the analog data. The GPS time does not take into account the earth's rotation. Corrections to the GPS time are made in the GPS receivers so that they provide UTC clock time. Loss of the time source may cause a bad time-stamp, angle drift, and frequency deviation (unless there is no back-up timing source occasional).

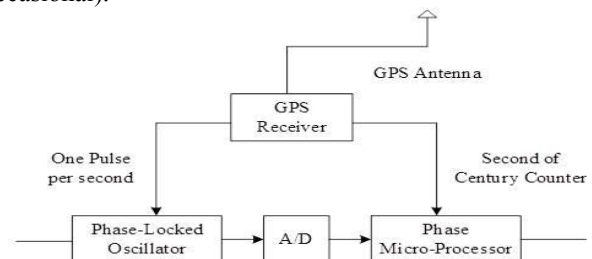


Figure 13. Indicator of the GPS loss and recovery flag

VII. CONCLUSIONS

Power system protection can be enhanced with the use of synchrophasor technology, starting with the use of PMU data for substation-level element protection schemes (since these avoid potential network latency problems). The next step is to ask whether the timing and quality of PMU data equal that of existing protection systems, and whether there is potential benefit to having PMU-based protection supplement relay-based protection measures. In-depth experimental trials of parallel PMU-based and relay-based protection systems should be studied to determine the potential benefits of synchrophasor-based protection.

Now that PMU data networks are more prominent and robust, it is vital that the quality of data be scrutinized prior to its use in applications. Data quality needs to be categorized and PMU data standards/registries need to be established and adhered to for reliable application. Current utility uses of PMU data have been limited to monitoring and analysis due to end user concerns over PMU data latency, reliability and questionable data quality, but these factors have been improving where they have been prioritized. Therefore, the focus in today's research should be on local substation protection schemes, and the first step is to compare and contrast existing relaying systems to PMUs for substation focused protection systems.

A new architecture founded on reimagining synchrophasor network transmission, routing and control lends itself to the advancement new protection functions with synchrophasors. For synchrophasor data collection (synchcollection), reduce latency by removing intermediate PDCs, place PMU data streams on dedicated/robust/redundant network paths and incorporate redundant GPS time sources through a time-distribution switch as proposed by Achanta et. al. through selection of best of diverse sources from GPS, IRIG-B, PTP and NTP [11]. For synchrophasor based protection (synchprotection) decisions, route signals through existing RTU control systems or setup a dedicated network of synchrophasor trip (synchtrip) devices at the substation level to receive trip signals from the synchrophasor decision controllers (synchcontroller).

The other immediate opportunity for synchrophasor use in system protection is to use PMU data to monitor and review every system protection operation, to determine whether each operation occurred as it should have, identify and diagnose any protection mis-operations, and identify potential equipment, settings or logic problems before they cause transmission or generation events.

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